12.3. Therapeutic Instruments

Pacemakers

Pacemaker is a medical device which uses electrical impulses, delivered by electrodes contracting the heart muscles, to regulate the beating of the heart

The primary purpose of a pacemaker is to maintain an adequate heart rate, either because the heart's natural pacemaker is not fast enough, or because there is a block in the hearts electrical conducting system.

The pacemaker system consists of pulse generator and a electrode. The pulse generator is a sealed box which contains lithium-halide cells to provide power and an electronic circuit to regulate the rate and pulse width of the electrical impulse.

This device uses low-energy electrical pulses to prompt the heart to beat at a normal rate. The pulse generator is placed under the skin below the collar bone. When required, the exhausted pulse generator can be replaced. Newer pacemakers also can monitor blood temperature, breathing, and other factors and adjust the heart rate.

Pacemaker batteries last between 5 and 15 years (average 6 to 7 years), depending on how active the pacemaker is. The wires of the pacemaker also may need to be replaced eventually.

Sick sinus syndrome (SSS) is characterized by dysfunction of the sinoatrial (SA) node. Treatment of SSS is directed at symptoms and typically involves the implantation of an artificial pacemaker.

Generally heart rate is low when resting. It's a sign of good health. When it is too low the condition is called **bradycardia**. Normally, the heart beats 60 to 100 times a minute when you are at rest. But with bradycardia, it goes down to less than 60 beats a minute. When the heart beat is too high it is called **tachycardia**

Medical LASER

Lasers are devices that emit a single, coherent wavelength of electromagnetic radiation that is used to cut, coagulate, or ablate tissue for a variety of clinical applications. The term "LASER" stands for **light amplification by stimulated emission of radiation**. Laser surgery uses non-ionizing radiation, so it does not have the same long-term risks as X-rays or other types of ionizing radiation. Laser systems produce a variety of wavelengths of varying pulse duration and energy levels.

Ordinary light, such as that from a light bulb, has many wavelengths and spreads in all directions. Laser light, on the other hand, has a specific wavelength. It is focused in a narrow beam and creates a very high-intensity light. Computer-based imaging and guidance systems allow procedures to be performed precisely, quickly, and with greater control. Although lasers are commonly used superficially for cutaneous and ocular applications, smaller efficient laser delivery systems are available for minimally invasive applications, including endoscopy, bronchoscopy, laparoscopy, and endovenous ablation.

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